

Melnick Nebart

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EDUCATION

Columbia University

New York, NY | 06.2027

Major: Astrophysics

Research Interests: Star formation, galactic dynamics, microgravity simulation, and space instrumentation design.

Relevant Coursework: Mechanics and Thermodynamics, Electromagnetism and Optics, Calculus 1, 2, Multivariable Calculus, Quantum and Mechanical Waves, Astrophysics Lab, Intro to Astrophysics I, Differential Equations, Linear Algebra

SKILLS

Computation: Python (NumPy, Matplotlib, SciPy, Pandas, Astropy, Photutils), CIAO (Chandra Interactive Analysis of Observations), LEGO Mindstorms Inventor, Microsoft Office, Solidworks, OpenSCAD

Research Methodology: Experimental Design, Qualitative and Quantitative Research Methods, Hypothesis Testing, Statistical Analysis, Data Collection and Analysis, Scientific Reporting

Analytical Skills: Data Interpretation and Visualization (Python: Matplotlib, Pyplot), Algorithm Optimization, Error Analysis, Stability and Convergence in Numerical Methods, and Performance Metrics Assessment.

Communication: Technical Writing, Poster Design, Peer Collaboration, Oral Presentation

HONORS AND AWARDS

Kluge Scholarship(\$100,000 per year)

2023, 2024, 2025

Columbia Dean's List

2024

Columbia Global Learner's Scholarship(\$16700)

2024

HERMAN FAMILY Summer Funding Award(\$3240)

2024

RESEARCH EXPERIENCE

Space Telescope Science Institute (STScI)

Baltimore, MD | 06.2025-09.2025

Research Intern [Supervisor: Rebecca Levy]

- Conducted star cluster identification in the starburst galaxy M82 using new Mid-Infrared Instrument (MIRI) data from the James Webb Space Telescope (JWST), where we detected 254 new embedded star clusters.
- Cross-matched multi-wavelength datasets (optical, near-infrared, submillimeter) to build a comprehensive census of star clusters in M82, contributing to studies on the full lifecycle of stellar cluster formation and evolution.

Columbia University: Research Skills

New York, USA | 01.2025-05.2025

Researcher [Supervisor: Alexandra Masegain, David Hefland]

- Retrieved and processed Gaia DR3 data for a designated stellar stream using Astroquery and Astropy, cleaning and validating positions, parallaxes, proper motions, radial velocities, and photometry.
- Modeled individual and ensemble stream orbits within a Milky Way potential using the Gala Python package, performing forward and backward integrations over 2 Gyr to assess progenitor convergence.
- Compared simulated stream morphologies from Plummer (star cluster) and NFW (dwarf galaxy) potentials against observed data, interpreting resultant kinematic and structural trends to infer the stream's origin.

Columbia Space Initiative

New York, USA | 09.2024-Present

Research Assistant (Engineering Focus)

- Collaborating on a research project investigating the effects of microgravity using a custom-built 3D clinostat, aiming to simulate space-like environments for biological and material science experiments.
- Contributing to the engineering design and construction of the 3D clinostat, optimizing rotational mechanisms to achieve microgravity conditions with an accuracy of $\pm 5\%$ for experimental simulations.
- Conducting performance tests and system integration assessments, enhancing device stability and precision, leading to a 20% improvement in replicating microgravity effects in biological samples.

Columbia Physics Scholars Program

New York, USA | 01.2024-05.2024

Research Assistant [Supervisor: John Staunton]

- Conducted experimental research on Dual Active Galactic Nuclei (AGN), identifying X-ray counterparts to gravitational wave sources, contributing to the ongoing study of supermassive black hole mergers.
- Analyzed and processed over 500 GB of data from the Chandra X-ray Observatory, utilizing Python and CIAO software to detect and categorize potential X-ray signals from AGN.

Vihiga Friends High School Sigma Club(Mathematics)

Vihiga, Kenya | 06.2019-02.2020

Researcher [Supervisor: Daniel Khaemba]

- Investigated advanced analytical methods for determining the radius of an inscribed circle in a triangle given two sides, utilizing principles from functional and numerical analysis to enhance algorithmic stability and convergence.
- Derived a new formula by integrating Heron's formula with geometric principles, achieving a 15% increase in computational efficiency for solving circle-related problems in irregular triangles.

- Presented findings to an audience of **100+** scientists and students at the Western Kenya Regional Science and Engineering Fair (KSEF) 2020, securing **2nd** place in the Mathematics category for innovative contributions to geometric analysis.

Vihiga Friends High School Robotics Program

Vihiga, Kenya | 01.2019-02.2020

Research Assistant [Supervisor: Samuel Simiyu]

- Led a research project on developing **autonomous navigation algorithms** for mobile robots in dynamic environments, improving robot obstacle detection and avoidance by **20%** using LEGO Mindstorms technology.
- Designed and implemented novel **motion planning algorithms**, integrating sensor-based navigation and path optimization techniques, enhancing robot trajectory accuracy by **15%**.
- Collaborated with a team of **5** classmates to refine the algorithms, presenting the project to an audience of 200+ at the **Kenya Regional Lego League Robotics Tournament**, securing 4th place at the Kenya Science and Engineering Fair 2020.

PUBLICATIONS AND PRESENTATIONS

Nebart, M., Levy, R. JWST Observations of Starbursts: A Multiwavelength Comparison of Young Massive Star Clusters in the Center of M82.

Research note in preparation, Space Telescope Science Institute, 2025.

Nebart, M. (2025). *Identifying Embedded Star Clusters in the Starburst Galaxy M82 Using JWST/MIRI*. Oral presentation delivered at the Space Telescope Science Institute (STScI) Summer Research Symposium, Baltimore, MD.

TEACHING EXPERIENCE

Department of Astronomy, Columbia University

New York, USA | 01.2025-05.2025

Teaching Assistant for ASTR UN1234: Universal Timekeeper [David Helfand]

- Graded 60+ student assignments and contributed to revising two lecture modules, improving clarity in course assessments.

ACTIVITIES AND ACADEMIC EXPERIENCES

Columbia University Housing

New York, USA | 01.2025-05.2025

Operations Assistant

- Managed and organized housing data in Asana, ensuring accurate record-keeping and efficient information access for the department.
- Collaborated with housing staff to streamline workflows, improving operational efficiency and enhancing resident satisfaction.
- Assisted in coordinating housing assignments, ensuring alignment with occupancy standards and student preferences.

Columbia Residential Life

New York, USA | 2025- Present

Residential Advisor

- Planned and facilitated educational, wellness, and social events, fostering a safe and supportive living environment for undergraduate students while promoting university values and community standards.
- Served in an on-call duty rotation, managing student concerns, responding to emergencies, and coordinating with professional staff to uphold residential policies and ensure student well-being.

NAFSA MIG United Nations

New York, USA | 12/2024- Present

Social Media Coordinator

- Served as Social Media Coordinator for the NAFSA NGO Youth Reps at the United Nations, leading the development and execution of digital strategy to amplify global education and migration-related initiatives.
- Collaborated with UN event organizers and NAFSA representatives to create and share timely content for high-level NGO events at the UN, fostering public engagement and awareness.

Columbia Scholastic Press Association

New York, USA | 03.2024, 2025

CSPA Redshirt Assistant

- Managed audio-visual coordination for a convention of 500+ attendees, troubleshooting technical issues to ensure seamless presentations.
- Organized logistics for 10+ events, achieving a 95% satisfaction rate from participants.
- Oversaw setup, staffing, and breakdown of registration stations, ensuring efficient operation throughout the event.

Columbia University Disability Services

New York, USA | 09.2023-12.2023

Transcriptionist

- Facilitated comprehensive note-taking support for 20+ students with disabilities in the Intro to Mechanics class, ensuring accessibility and inclusivity in academic materials.
- Received commendations from faculty for responsiveness and exceptional communication, leading to improved support services.

Campus Reels

New York | 2023

Content Creator

- Created engaging short-form video content showcasing student life, academics, and campus culture at Columbia University for prospective students.
- Scripted, filmed, and edited original video content that aligned with CampusReel's brand and storytelling goals.
- Promoted authentic student perspectives to enhance college access and transparency for high school applicants from diverse backgrounds.

Equity Bank Limited

Ongata Rongai, Kenya | 05.2022-08.2022

Financial Operations Intern

- Processed cash system transactions for 100+ customers daily, ensuring accurate and efficient financial operations.
- Computed financial records and monitored inventory levels for up to 100 customers per day.
- Improved customer service rankings by 3%, achieving a 90% satisfaction rate through effective service delivery.

Columbia Global Centre Nairobi

Nairobi, Kenya | 07.2024-09.2024

Programs Assistant Intern

- Coordinated logistics for 15+ events, including venue booking and catering, enhancing event efficiency and attendee satisfaction including the Kenya Space Education Summit
- Conducted in-depth research on 5+ relevant topics, informing program development and enhancing community outreach efforts.
- Designed and distributed promotional materials, increasing program visibility and attendance by 20%.

LEADERSHIP AND OUTREACH EXPERIENCES

Equity Global Scholars Board

New York, USA | 01.2024 - Present

First Year Representative | USA Representative

- Represented 100+ first-year Equity Global Scholars, facilitating engagement and mentorship initiatives to enhance community integration and support.
- Provided mentorship to 15 incoming scholars, aiding in their transition to college and academic success.

Kenya Space Education Summit

Nairobi, Kenya | 07.2024-08.2024

Co-Coordinator

- Spearheaded the **Kenya Global Space Education Summit**, coordinating over 10 scientific presentations and workshops on space science and satellite technology, directly impacting the education of 200+ students and educators.
- Analyzed post-event data from **150+** participants, producing a comprehensive report that measured a **30%** increase in knowledge retention related to space science topics, contributing to ongoing improvements in future educational initiatives.

Equity Leaders Program- Kenya

04.2023- Present

College Counselling Consultant

- Collaborated with 35 mentors to enhance the mentoring program, incorporating feedback that improved participant engagement by 30%.
- Mentored **10 high school** graduates from Kenya, Uganda, Rwanda, and the DRC in their college applications, achieving a **90% acceptance rate** while enhancing leadership, communication, and critical thinking skills through organized workshops and networking events.

ADDITIONAL INFORMATION

Languages: English (Fluent), Swahili (Fluent), French (Elementary)

Workshops: Chandra/CIAO Workshop 2025, Columbia Undergraduate Mathematical Modeling Workshop (CUMMW)

Memberships: National Society of Black Physicists(2024- Present), Society of Physics Students(2024-present)

Interests: African Traditional Music, Jazz, Drumming, Science fiction, Memes, cooking international cuisine, visiting museums, basketball